26 January 1973
SUPERSEDING
MIL-C-55639(EL)
26 April 1969

MILITARY SPECIFICATION

CHARGER, BATTERY PP-4127( )/U

1. SCOPE

1.1 This specification covers a solid state, portable battery charger designated as Charger, Battery, PP-4127( )/U for charging 6 volt, vented, zinc silver oxide batteries. (See 6.1 and 6.3)

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-781	Strapping, Steel, Flat And Seals
PPP-B-585	Box, Wood, Wirebound
PPP-B-601	Boxes, Wood, Cleated-plywood
PPP-B-621	Box, Wood, Nailed And Lock-corner
PPP-B-636	Box, Fiberboard
PPP-B-640	Box, Fiberboard, Corrugated, Triple-Wall
PPP-F-320	Fiberboard, Corrugated And Solid, Sheet Stock (Container Grade) And Cut Shapes
PPP-S-760	Strapping, Nonmetallic, (And Connectors)
PPP-T-76	Tape, Pressure-Sensitive Adhesive Paper, Water Resistant, (For Carton Sealing)
PPP-T-97	Tape, Pressure-Sensitive Adhesive, Filament Reinforced

MILITARY

MIL-P-116	Preservation, Methods Of
MIL-T-152	Treatment, Moisture And Fungus-resistant, Of Communications, Electronic, And Associated Electrical Equipment
MIL-P-3938	Pallet, Material Handling, Hardwood, Stringer Construction, 4-way (Partial)

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MIL-P-11268	Parts, Materials, And Processes Used In Electronic Equipment
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finish For Ground Signal Equipment
MIL-P-15011	Pallet, Material Handling, Wood, Post Construction, 4-way Entry
MIL-P-55110	Printed Wiring Boards

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures And Tables For Inspection By Attributes
MIL-STD-129	Marking For Shipment And Storage
MIL-STD-147	Palletized And Containerized Unit Loads 40 Inch x 48 Inch Pallets, Skids, Runners, Or Pallet Type Base
MIL-STD-252	Wired Equipment, Classification of Visual And Mechanical Defects
MIL-STD-275	Printed Wiring For Electronic Equipment
MIL-STD-454	Standard General Requirements For Electronic Equipment
MIL-STD-461	Electromagnetic Interference Characteristics Requirements For Equipment
MIL-STD-462	Electromagnetic Interference Characteristics Measurement Of

DRAWINGS

ELECTRONICS COMMAND

DL-SM-A-610293	Charger, Battery PP-4127( )/U
DL-SM-A-610296	Battery Charger Subassembly
DL-SM-A-610297	Control Circuit Board Assembly
SM-B-601491	Test Specification Battery Charger
SM-B-601492	Test Specification Control Circuit Board
SM-B-609932	Test Procedure for Battery Charger
SM-B-609934	Test Procedure for Control Circuit Board Assembly
SM-B-610008	Conformal Coating (Epoxy), Application of
SC-A-678001	Burn-in/Debugging Procedure for Charger, Battery PP-4127( )/U
SM-A-686335	Specification, Environmental Battery Charger

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies.)

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3. REQUIREMENTS

3.1 Preproduction samples. - The contractor shall furnish three preproduction samples of Charger, Battery PP-4127( )/U for approval, if required by the invitation for bids and contract. (See 4.3 and 6.2(d)(1))

3.2 Construction. - The equipment shall be constructed in accordance with the requirements of this specification, and of the following drawings (including the requirements for parts, materials, and processes thereon): (See 4.9)

DL-SM-A-610293 Charger Battery PP-4127( )/U
DL-SM-A-610296 Battery Charger Subassembly
DL-SM-A-610297 Control Circuit Board Assembly

3.3 Parts, materials, and processes; general. - In addition to the requirements of this specification, the requirements of MIL-P-11268, including the selection requirements therein, shall apply. (See 6.8)

3.4 Printed wiring assemblies. - (See 3.3 and 4.4)

3.4.1 Printed wiring boards. - Printed wiring boards used in the construction of printed wiring assemblies shall be a product which has passed the requirements of MIL-P-55110, MIL-STD-275, and the master drawing. Preproduction test requirements per MIL-P-55110 shall be performed as specified.

3.4.2 Protective coating. - Printed wiring assemblies shall be protected on both sides of the printed wiring assembly with an epoxy-type conformal coating to protect the pattern against moisture, dust and other contaminants. The coating shall be in accordance with Drawing SM-B-610008.

3.5 Finish, protective. - Equipment shall be given protective finish in accordance with MIL-F-14072. This includes finish of hardware, such as handles, hinges, screws, etc., and necessary touch up after mounting. The final paint film on Type I surfaces shall be green color (olive drab), semi-gloss enamel, matching a color chip provided by the procuring agency. (See 4.4 and 6.6)

3.6 Marking. - Marking shall conform to MIL-M-13231. Front panel marking shall be Group I as described in that specification. (See 4.4)

3.6.1 Visibility. - Whenever practicable, parts and assemblies shall be so mounted that their identification markings will be readily visible with minimum disassembly of the equipment.

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3.6.2 Serial numbers. - Each Charger, Battery PP-4127( )/U shall be serial numbered.

3.7 Electrical requirements. -

3.7.1 Battery charger. - The battery charger shall meet the requirements specified in Drawing SM-B-601491. (See 4.7.1)

3.7.2 Printed wiring assembly. - The printed wiring assembly shall meet the requirements specified in Drawing SM-B-601492. (See 4.7.2)

3.8 Service conditions. - The equipment shall meet the following service conditions:

3.8.1 Operation (general). - The equipment shall operate under environmental service conditions specified in Drawing SM-A-686335 with no more than normal maintenance and replacement of parts. Where the contractor is required to make a selection of parts, materials, processes, construction methods, etc., he shall be guided by this requirement and 6.8. Approval of the preproduction sample (see 3.1) shall be considered as compliance with this requirement.

3.8.2 Service condition operational requirements. - Prior to, during, and after testing as specified for each of the service conditions, the equipment shall conform to the requirements of paragraph 5 of Drawing SM-B-601491. (See 3.7.1 and 4.8)

3.9 Overall treatment (moisture and fungus proofing - treatment (MFP)). - Moisture and fungus proofing treatment shall be in accordance with MIL-T-152. The equipment (including accessories) shall show no evidence of viable fungus or corrosion when subjected to the fungus test specified in 4.8. Corrosion is any visible degradation that can be attributed to flaky, pitted, blistered, or otherwise loosened surfaces.

3.10 Electromagnetic interference characteristics. - The equipment when operated and loaded as specified herein shall meet the requirements of CE04, CE05 and RE02 of MIL-STD-461A Notice 4. Test and test methods shall be as specified in 4.10.

3.11 Burn-in/Debugging preconditioning. - The equipment shall meet the requirements herein, without subsequent processing, after subjection to the burn-in period of 4.6. (See 4.5)

3.12 Operational test. - When subjected to the test specified in 4.11, the equipment shall meet the requirements of paragraph 5 of Drawing SM-B-601491. (See 3.7.1 and 6.7)

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3.13 Air-seal test. - The transit case shall provide means for enabling performance of the air-seal test. (See 4.12 and 6.7)

3.14 Interchangeability. - Like units, assemblies, subassemblies and replaceable parts shall conform to requirement 7 of MIL-STD-454. (See 4.13)

3.15 Technical literature, tools, and running spare parts. - Technical literature, tools, and running spare parts shall be furnished as specified by the contract. Running spare parts shall be identical to corresponding parts in the equipment furnished on order. (See 6.2(g) and 6.2(h))

3.16 Workmanship. - The equipment shall be manufactured and assembled in accordance with requirement 9 of MIL-STD-454 and the applicable portions of the following paragraphs:

(a) In MIL-P-11268:

(1) General requirements for plastic material and plastic parts.

(2) Wiring and cabling, including: Slack, protection, clearance, splicing, connections in general, grounding, shielding on wire and cable.

(3) Flux and cleaning agents for solder.

(4) Soldering.

(5) Cleaning.

(6) Riveting.

(7) General requirements for securing of parts.

(8) Welding.

(9) Tropicalization.

(10) Mechanical operation of controls.

(11) Self-tapping screws.

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4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection. - Inspection shall be classified as follows:

(a) Preproduction inspection (does not include preparation for delivery). (See 4.3)

(b) Inspection covered by subsidiary documents. (See 4.4)

(c) Quality conformance inspection.

(1) Quality conformance inspection of equipment before preparation for delivery. (See 4.5)

(2) Quality conformance inspection of preparation for delivery. (See 4.15)

4.3 Preproduction inspection. - This inspection shall consist of the inspection specified in the subsidiary documents covering the items listed in 4.4, and the inspection specified for group A, group B, and environmental/design (See table II, III and I respectively) after subjection to burn-in/debugging. (See 3.11). Preproduction inspection shall be performed on all samples except for environmental/design inspection which shall be performed on the samples specified and in the sequence shown in Table I.

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automatically by means of a thermostatic valve.

Table I. - Environmental/Design inspection

Inspection	Req. Para.	Insp. Para.
Sample No. 1		
Low temperature	3.8.2	4.8
High temperature	3.8.2	4.8
Temperature shock	3.8.2	4.8
Rain	3.8.2	4.8
Humidity (moisture resistance)	3.8.2	4.8
Sample No. 2		
Altitude	3.8.2	4.8
Orientation	3.8.2	4.8
Electromagnetic interference	3.10	4.10
Salt fog	3.8.2	4.8
Sample No. 3		
Leakage (immersion) - See Note	3.8.2	4.8
Dust	3.8.2	4.8
Vibration	3.8.2	4.8
Bounce (vibration)	3.8.2	4.8
Shock, drop	3.8.2	4.8
Bench handling	3.8.2	4.8
Fungus	3.8.2	4.8

NOTE: The leakage (immersion) inspection shall be performed at the beginning of the test sequence, and again after completion of the shock, drop inspection.

4.4 Inspection covered by subsidiary documents. - The following shall be inspected under the applicable subsidiary documents as part of the inspection before preparation for delivery:

Item	Where Required
Printed Wiring Assemblies	3.4 and 3.7.2 (See Note A)
Finish	3.5
Marking	3.6

NOTE A: Each printed wiring assembly shall be tested in accordance with 4.7.2 prior to installation.

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4.5 Quality conformance inspection of equipment before preparation for delivery. - Each unit of each lot of equipment shall be subjected to burn-in/debugging (See 3.11) prior to inspection. The contractor shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition the Government--at its discretion--may perform all or any part of the specified inspection, to verify the contractor's compliance with specified requirements. (See 6.4). Test Equipment for Government verification inspection shall be made available by the contractor.

4.5.1 Group A inspection. - Each unit of each lot of equipment shall be inspected for conformance to all the inspection and test requirements of Table II. Each lot will be subject to verification, utilizing the procedures of MIL-STD-105 using the general inspection levels;and the AQL's indicated in Table II. Group A inspection shall be performed in any order which is satisfactory to the Government, except that the operational inspection (4.11) shall be next to last and the air-seal test (4.12) shall be last. (See 6.7)

Table II. - Group A inspection

Inspection	Req. Para.	Insp. Para.	AQL Major	AQL Minor
Visual and mechanical	3.16	4.14	2.5 DPHU *	10.0 DPHU *
Electrical tests				
Input voltage	3.7.1	4.7.1	1.5 DPHU For The Group	**
Charging voltage	3.7.1	4.7.1		
Output current regulation	3.7.1	4.7.1		
Voltage cutoff	3.7.1	4.7.1		
Operational test	3.12	4.11	0.1%	**
Air Seal	3.13	4.12	1.0%	**

* DPHU - DEFECTS PER HUNDRED UNITS
** ALL DEFECTS ARE CONSIDERED MAJOR

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4.5.2 Group B inspection. - This inspection, including sampling, shall conform to Table III and to special procedures for small-sample inspection of MIL-STD-105. Group B inspection shall be performed on inspection lots that have passed group A inspection and on samples selected from units that have been subjected to and met group A inspection.

Table III. - Group B inspection

Inspection	Req. Para.	Insp. Para.	AQL
Electrical tests			
Duty cycle	3.7.1	4.7.1	4.0%
Overload protection	3.7.1	4.7.1	4.0%
Undervoltage protection	3.7.1	4.7.1	For the
Power Factor and Power Dissipation	3.7.1	4.7.1	Group
Interchangeability	3.14	4.13	6.5%

4.5.2.1 Group B sampling plans. - The group B sampling plans, for the AQL's listed in Table III, shall be as follows:

AQL	Inspection Level
4.0%	S-4
6.5%	S-4

4.5.2.2 Order of inspection within group B. - Group B inspection shall be performed in any order which is satisfactory to the Government.

4.5.3 Group C inspection. - This inspection shall consist of the tests listed in Table IV and shall be performed on sample units that have been subjected to and met Groups A and B inspection. Sample units shall be selected in accordance with 4.5.3.1.

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be furnished to Ships [illegible] shall be furnished. One set of detail drawings shall

Table IV. - Group C inspection

Inspection	Req. Para.	Insp. Para.
Group I		
Altitude	3.8.2	4.8
Low temperature	3.8.2	4.8
High temperature	3.8.2	4.8
Temperature shock	3.8.2	4.8
Humidity (moisture resistance)	3.8.2	4.8
Group II		
Electromagnetic interference	3.10	4.10
Vibration	3.8.2	4.8
Bounce (vibration)	3.8.2	4.8
Shock, drop	3.8.2	4.8
Leakage, (Immersion)	3.8.2	4.8
Salt Fog	3.8.2	4.8
Group III		
Dust	3.8.2	4.8
Rain	3.8.2	4.8
Fungus	3.8.2	4.8
Group IV		
Orientation	3.8.2	4.8
Bench handling	3.8.2	4.8
Group V		
Equipment Verification Review	3.2	4.9

4.5.3.1 Sampling for Group C inspection. -

4.5.3.1.1 Group I and Group II - Table IV. - One (1) sample shall be selected at random for each group, from each 200 equipments produced, and subjected to the inspection specified, in the order shown. The first samples shall be selected at the start of the contract from the first quality conformance inspection production lot. (See 4.5.3)

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4.5.3.1.2 Group III - Table IV. - One (1) sample shall be selected at random from each 400 equipments produced and subjected to the inspections specified, in the order shown. The first sample shall be selected at the start of the contract from the first quality conformance inspection production lot. (See 4.5.3)

4.5.3.1.3 Group IV - Table IV. - One (1) sample shall be selected at random from each 100 equipments produced and subjected to the inspections specified. The first sample shall be selected at the start of the contract from the first quality conformance inspection production lot. (See 4.5.3)

4.5.3.1.4 Group V - Table IV. - The Equipment Verification Review (EVR) shall be performed on the initial production lot. No subsequent inspection for EVR will be required. (See 4.9)

4.5.3.2 Noncompliance. - The contractor shall immediately report in writing each group C failure occurrence, including details of the failure and characteristics affected. The contractor shall immedaitely investigate the cause of failure and further report the results of investigation and details of the proposed corrective action on (1) the process and materials, as applicable, and (2) all units of product which were manufactured under the same conditions which the Government considers subject to the same failure. Reports shall be forwarded to the responsible technical activity designated in the contract through the Quality Assurance Representative. After corrective action has been taken, additional sample units shall be subjected to group C inspection (all inspections, or the inspections which the sample failed, at the option of the Government) and groups A and B inspection may be reinstitutec; however, final acceptance and shipment will be withheld until the group C reinspection results have shown that the corrective action was effective. (See 6.5)

4.5.4 Reinspection of conforming Group B and Group C sample units. - Unless otherwise specified, sample units which have been subjected to and passed group B and group C inspection may be accepted on the contract or order provided all damage is repaired and the sample units are resubjected to and pass group A inspection.

4.6 Burn-in/Debugging procedures. - Each equipment shall be subjected to the preconditioning period specified in Drawing SC-A-678001. After completion of the preconditioning period, the unit shall not be repaired, alined or otherwise changed prior to subjection to acceptance inspection. (See 3.11 and 4.5)

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4.7 Electrical tests.-

4.7.1 Battery charger. - The battery charger shall be tested in accordance with Drawing SM-B-609932 to determine compliance with the requirements of 3.7.1.

4.7.2 Printed wiring assembly. - The printed wiring assembly shall be tested in accordance with Drawing SM-B-609934 to determine compliance with the requirements of 3.7.2. (See 4.4)

4.8 Service condition tests. - Charger, Battery PP-4127( )/U shall be tested in accordance with Drawing SM-A-686335 to determine compliance with the requirements specified in 3.8.2.

4.9 Equipment Verification Review. -

4.9.1 The contractor shall perform an Equipment Verification Review (EVR), consisting of a complete technical audit of the equipment on order against the drawings cited for construction (par. 3.2). The EVR shall consist of the following:

4.9.2 An audit to establish that "as-built" equipment (unit, assembly, module, or part) is in accordance with the cited end product drawings.

4.9.3 An audit to establish that "as-built" configuration of unit, assembly, module, or part meet the acceptance requirements (other than equipment specification performance) specified for each unit, assembly, module, or part. The audit shall record all acceptance test methods used together with resulting verification test data.

4.9.4 The contractor shall compile an EVR report (s) containing the following information:

(a) Identification of unit, assembly, module, or part and details proving compliance with 4.9.2 and 4.9.3 above.

(b) Discrepancies noted.

(c) Corrective action taken.

4.9.5 The contractor shall make available to the Government copies of his EVR report(s) no later than 15 days prior to the submission of the first production unit or lot. The results of the EVR shall be subject to inspection by authorized Government personnel at the time the first unit or production lot is offered for acceptance. Government inspection will be to the depth and extent necessary to demonstrate that the "as-built" hardware is in accordance with the cited drawings. Copies of the EVR reports shall be made available to the Government personnel during the inspection. The contractor shall provide the following:

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(a) Segregation of units, assemblies, modules, and parts to permit reviewing personnel access for detailed inspection.

(b) As required, responsible personnel from each functional department available for discussions in their respective areas.

(c) Adequate administrative support for the EVR.

4.10 Electromagnetic interference characteristics. - Conformance of the test sample in accordance with 3.10 shall be determined by the contractor or an approved testing laboratory. Testing shall be in accordance with MIL-STD-462 Notice 3. The test item shall be operated using a 115 vac, 60 Hz input and an output load consisting of battery BB-622. Prior to production, the contractor shall furnish the contracting officer, for approval, a test report which shall include a block diagram of the test set-up and a list of the test equipment utilized. Upon approval by the contracting officer, and provided all other requirements of this specification are met, the approved test sample shall be used as a model for visual comparison of production units.

4.11 Operational test. - With an input of 115 vac, 60 Hz, the equipment shall meet the requirements specified in 3.12. (See 4.5.1 and 6.7)

4.12 Air seal test (See 3.13). - The equipment shall be opened and closed again in such manner as to break and remake the seal. Immediately thereafter, the equipment as field transported shall be subjected to a vacuum of 1 pound per square inch (1 pound per square inch less than the atmospheric pressure surrounding the equipment) applied to the interior of the transit case or to the interior of the equipment inclosure. The vacuum then shall be valved-off and the interior pressure measured during the ensuing period of 1 minute. During this 1-minute period the decrease in vacuum shall not exceed 0.01 pound. The gage used for measurement of the vacuum shall be of such accuracy that a difference of 0.01 pound can be determined readily. Extreme care shall be exercised to assure that no foreign matter will enter the unit under test. (See 4.5.1 and 6.7)

4.13 Interchangeability. - The dimensions listed below shall be measured to determine conformance to the physical interchangeability requirement of 3.14. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

(a) External and internal dimensions of cases, covers and insertable assemblies; when such dimensions affect mating of parts.

(b) Dimensions of cavities, when such dimensions affect insertion of items.

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(c) Location of hinges and fasteners on separable parts or assemblies which must mate, such as cases, covers and mountings.

(d) Location of connectors, locking pins, fasteners, slides, and mountings which receive mating parts of plug-in assemblies and major units, and location of the mating parts on the plug-in assembly or major unit.

(e) Size and form of special threads.

4.14 Visual and mechanical inspection. - The equipment shall be examined for the defects listed in MIL-STD-252. (See 3.2 and 3.16)

4.15 Quality conformance inspection of preparation for delivery. - Preparation for delivery shall be inspected in accordance with Specification MIL-P-116 to determine conformance to the requirements of Section 5.

4.16 Rough handling test (preparation for delivery). - When rough handling test in accordance with MIL-P-116 is required by contract (See 6.2(d)(2)(c)), the following functional test shall be conducted to determine freedom from operational malfunction caused by the rough handling:

Insp. Par.	Measurement
4.11	Output and voltage cutoff

5. PREPARATION FOR DELIVERY

5.1 Preservation and Packaging. - Preservation and packaging shall be level A or C as specified. (See 6.2(b))

5.1.1 Level A. -

5.1.1.1 Cleaning. - Each Charger, Battery PP-4127( )/U shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. - Each Charger, Battery PP-4127( )/U shall be dried in accordance with the applicable procedure of MIL-P-116.

5.1.1.3 Preservation application. - None required.

5.1.1.4 Unit packaging. - Unit packaging shall be in accordance with the methods prescribed in MIL-P-116 as specified herein.

5.1.1.4.1 Technical literature. - Each technical literature shall be packaged method IC-1.

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5.1.1.4.2 Charger, Battery PP-4127( )/U. - Each charger shall be packaged method III as follows: cushion each charger on all surfaces with cells or pads, or both, fabricated of fiberboard conforming to PPP-F-320, W5C, weather resistant, designed and fabricated to protect all projections and absorb the shock of impact in handling and transit. Place the cushioned item within a close-fitting fiberboard box conforming to PPP-B-636, W5C, weather resistant. Place the technical literature packaged as specified in 5.1.1.4.1 in the box directly under the lid of the box. Box closure will be as specified in the appendix of the box specification.

5.1.2 Level C. - Each Charger, Battery PP-4127( )/U shall be preserved and packaged in a manner that will afford adequate protection against physical and environmental damage during shipment, handling and limited intransit storage.

5.2 Packing. - Packing shall be level A, B, or C as specified. Shipping containers for all levels shall be capable of stacking and supporting superimposed loads, during shipment and storage without damaging the container(s) or its contents. (See 6.2(b))

5.2.1 Level A. -

5.2.1.1 Palletized load. - A quantity of chargers, packaged as specified in 5.1, shall be placed on a pallet, load type 1, conforming to MIL-STD-147. A fiberboard cap shall be employed over the load having two sides extending down the stacked load at least 12 inches to accommodate marking requirements. The cap shall be fabricated of fiberboard conforming to PPP-F-320, class weather-resistant, W5s or V3c. The load shall be "bonded" to the pallet by strapping.

5.2.1.2 Less than palletized load. - When quantities per destination are less than a pallet load, the item packaged as specified in 5.1, shall be waterproofed, with tape conforming to PPP-T-76, in accordance with the taping requirements of the appendix of the box specification. A quantity of the waterproofed containers shall be placed within a close-fitting box conforming to PPP-B-601, overseas type; PPP-B-621, style 4, class 2; or PPP-B-585, style 2 or 3, class 3. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, 3 x 4 inch skids, laid flat, shall be applied in accordance with the requirements of the container specification, or if not specified in the specification, in a manner which will adequately support the item and facilitate the use of material handling equipment. Closure and strapping shall be in accordance with the applicable container specification or appendix thereto except that metal strapping shall conform to QQ-S-781, type 1, class B.

5.2.2 Level B. -

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IN REPLY

5.2.2.1 Palletized load. - A quantity of chargers, packaged as specified in 5.1, shall be palletized as specified in 5.2.1.1.

5.2.2.2 Less than palletized load. - When quantities per destination are less than a pallet load, a quantity of chargers, packaged as specified in 5.1, shall be placed within a close-fitting fiberboard box conforming to PPP-B-640, class 2, style E, or PPP-B-636, type CF, class weather-resistant, variety DW. The gross weight of boxes conforming to PPP-B-640 shall not exceed 250 pounds. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, containers will be pallet-mounted on pallets conforming to MIL-P-15011 or MIL-P-3938. Closure shall be in accordance with the appendix of the applicable box specification. Reinforcing shall be by nonmetallic strapping or pressure-sensitive filament tape banding conforming to PPP-S-760, type II and PPP-T-97, type IV respectively; selection of the material and application shall be in accordance with the appendix of the applicable box specification.

5.2.3 Level C. -

5.2.3.1 Palletized load. - A quantity of chargers, packaged as specified in 5.1, shall be palletized as specified in 5.2.1.1 except that the fiberboard caps shall be class domestic.

5.2.3.2 Less than palletized load. - When quantities per destination are less than a pallet load, a quantity of chargers, packaged as specified in 5.1, shall be packed as specified in 5.2.2.2 except that the fiberboard boxes shall conform to PPP-B-640 and PPP-B-636, class 1 and class domestic, respectively, and reinforcing shall not be required for boxes conforming to PPP-B-636.

5.3 Marking. - In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. - Battery Charger PP-4127( )/U is intended for charging two BB-622, 6V zinc silver oxide batteries simultaneously. These batteries are used with Radar Set AN/PPS-5. (See 1.1)

6.2 Ordering data. - Procurement documents should specify the following:

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(a) Title, number, and date of this specification and any amendments thereto.

(b) Level of packaging and level of packing required for shipment. (Level A, level B, or level C)

(c) The specific paragraphs of Section 5 which are applicable to this procurement.

(d) Preproduction inspection:

(1) Three sample units of each item cited in section 1 are generally required so that lengthy and potentially damaging environmental tests can be performed on several sample units. (See 3.1)

(2) Preproduction pack(s) as follows:

(a) Makeup of pack(s).

(b) Number of each kind of pack to be submitted.

(c) Inspection to be performed thereon. (See 4.16)

(e) Marking and shipping of samples.

(f) Place of final inspection.

(g) Technical literature required. (See 3.15)

(h) Quantity of tools and running spare parts required. (See 3.15)

(i) Submission of the statement of treatment covering tropicalization of material as soon as possible after award of contract. This statement should be submitted to the contracting officer. (See 3.10.3)

6.3 Nomenclature. - The parenthesis in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: PP-4127W/U. The contractor shall apply for nomenclature (Type Designation) in accordance with the applicable clause in the contract. (See 1.1)

6.4 Verification inspection. - Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

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6.5 Group C inspection. - Approval to ship may be withheld, at the discretion of the Government, pending the decision from the contracting officer on the adequacy of the corrective action. (See 4.5.3.2)

6.6 Color. - The color chip furnished by the procuring agency (3.5) will match color chip no. X-24087 may be obtained upon request to Commanding General, Coating and Chemical Laboratory, Aberdeen Proving Ground, Maryland.

6.7 Location of operational test and air-seal test. - It is desirable that the operational inspection (4.11) and the air-seal test (4.12) be performed at a location that will minimize handling (which might cause damage to the equipment) after this inspection is completed. Any preparation for shipment which would require breaking of the equipment seal should be accomplished prior to the air-seal test so that the seal will remain intact thereafter. It is recommended that the entire lot (including all previously inspected sample units) be inspected immediately prior to packaging. (See 4.5.1)

6.8 Qualitative material requirements. - The following service conditions were considered in the design of the equipment and are given here as a guide for processes, construction methods, etc.

6.8.1 Temperature. -

(a) Operating: Ambient temperature in the range of + 155°F to - 40°F (The high temperature includes effect of sun-load). Exposure at the low temperature not to exceed 6 hours, at any one time.

(1) Heat dissipation. - The internal ambient temperature and temperature of any component when the battery charger is operating at an external ambient of plus 155 degrees fahrenheit (unless otherwise specified herein) shall not exceed the limit established for any component contained in the battery charger after all internal temperatures have stabilized.

(2) Cooling. - Safe operating equilibrium temperature shall be obtained for the battery charger over the specified ambient temperature range, minus 40 degeees fahrenheit to plus 155 degrees fahrenheit, through the use of convection cooling. Exposed parts, including the inclosures of the equipment, shall not achieve a temperature in excess of 140 degrees fahrenheit when operating at an ambient temperature of 77 degrees fahrenheit. The temperature of the front panel and operating controls shall not exceed 109 degrees fahrenheit at the same ambient temperature.

(b) Nonoperating: Exposure in the range of + 160°F to - 65°F; exposure at the high temperature extreme not to exceed 4 hours, and at the low temperature extreme not to exceed 6 hours, at any one time.

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FSC remarks of SF1403 are addressed in QA survey report and are positive. Although the QA Manager

6.8.2 Relative humidity. - Up to 97 percent relative humidity for 20 hours; and exposure at 100 percent relative humidity, with condensation, for 4 hours.

6.8.3 Altitude (elevation.) -

(a) Operating: Up to 10,000 feet above sea level.

(b) Nonoperating: Up to 40,000 feet above sea level.

6.8.4 Orientation. -

(a) Operating: Any orientation from normal operating position (that is: forward, backward, left, or right.)

(b) Nonoperating: Storage in any position for a period of two years.

6.8.5 Immersion. - No leakage when placed in three feet of water for 2 hours.

6.8.6 Rain. - Two inches of rainfall per hour, with wind at 40 miles per hour.

6.8.7 Dust. - As encountered during heavy traffic over dry, clay, roads or in desert areas.

6.8.8 Sea salt fall-out. - Salt fall-out as experienced on the coast near warm bodies of water.

6.8.9 Stability. - In relation to specified input, output, charge mode, and temperature, the closed or open loop control system used, or combination thereof, shall not exhibit an unusual mode of operation. This stability shall also include the condition of no load, battery disconnected.

CUSTODIAN:

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PREPARING ACTIVITY

Army-EL

Project Nr. 6130-A127

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-C-55639A(EL) CHARGER, BATTERY PP-4127( )/U

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

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